AME 294 · PG3: UNREAL OR UNITY MCP GAME

MISSION: APEX STRIKE

A magnetic-boots puzzle game in Unity 6, and how it was built with AI

William Edwards



THE GAME

WALK ON WALLS. DROP ON PURPOSE.

A 3D puzzle obstacle course. Magnetic boots stick to bright panels, curves carry you onto walls and ceilings, and switching the boots off drops you back to the floor.

The original brief: *“an obstacle course with gravity effects enabled by Magnetic Boots… sci-fi/Portal style.”*





MECHANICS

FOUR RULES, COMBINED ROOM BY ROOM



Curves turn you

Quarter-pipes carry you onto walls and ceilings. A sharp corner is a dead end.



Drops are the puzzle

Boots off: fall and land upright. A marker shows where: blue safe, red void, gold sentry.



Sentries see a plane

One hit kills and speed won't save you. Drop on the dome to destroy one.



Timing is exact

Platforms and shuttles run on a clock that rewinds on respawn, so every try is identical.

THE CHAMBER

FOURTEEN ROOMS, ONE IDEA AT A TIME

Learn · rooms 1–10

Walk · Boots · First Curve · Ceiling · Over the Edge · Island · Platforms · Shuttle · Sentry · Stomp

Each room introduces one idea, with a checkpoint after it.

Combine · rooms 11–13

Finale · Corkscrew · Diving Board

Floor → wall → ceiling → wall, then a drop from a wall onto a sentry; a jump onto a ferry.

Challenge · room 14

Gauntlet

Two ceiling shuttles meet for 2 s; drop onto an island sentry; a running jump onto a second sentry.

Every room is proved solvable: a scripted run finishes the whole chamber with zero deaths, in about 225 s.

HOW IT WAS BUILT

BY THE NUMBERS

170
prompts to the coding agent, all logged

344
automated tests (62 EditMode, 282 PlayMode)

17 → 6
concept images, turned into six Tripo models

≈750
Tripo credits, including drafts and retries

31 → 13
sound generations, narrowed to the 13 in the game

14
rooms, proved solvable by a scripted run

TOOLCHAIN

ONE AGENT DRIVING THE ENGINE

Claude Code

Claude Opus 5.5

All code and tests, level building, shader, audio wiring, UI, builds

Unity 6

URP · CLI · Pipeline

The agent ran C# in the open Editor, Play mode, tests and captures

Codex CLI

GPT image tool

17 concept images, called by Claude Code

Tripo Studio

Pro plan

Six models, auto-rig, eight animation clips

ElevenLabs

Creator plan

13 sound effects through the API; two music tracks

The agent didn’t just write files: it controlled the running Editor, so it could build levels, play them and check the result itself.

PROCESS

THE LOOP BEHIND EVERY FEATURE

Describe
I set the goal and the constraints

→

Plan
The agent proposes; I question and decide

→

Build
Code and tests, then it plays the result in the Editor

→

Playtest
I play it and say what feels wrong

→

Commit
Only on my go-ahead

Playtest notes loop back to Build until it feels right. White steps are mine; blue steps are the agent’s.

Grey boxes first. Art and sound went onto a game that already played well.

Measure, then fix. Logs, captures and sampled animation before any change.

Test every orientation. Each movement rule checked on floors, walls and ceilings.

3D ASSET WORKFLOW

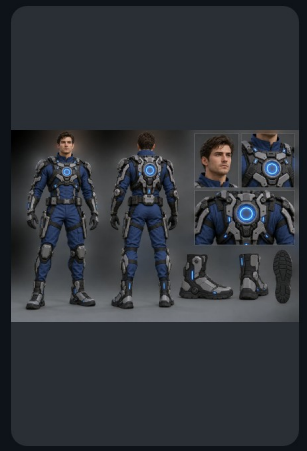
FROM A SENTENCE TO A PLAYABLE MODEL

1 · Prompt

"Athletic man in blue jumpsuit with sci-fi exoskeleton, glowing blue circle on chest..."

I describe the asset

2 · Codex image



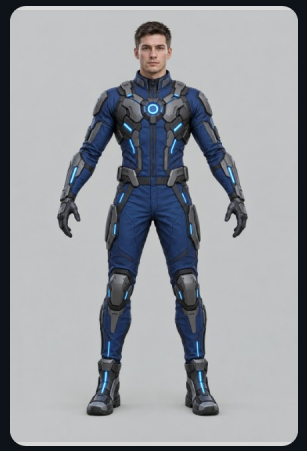
Good design, busy sheet

3 · Refined

One view · A-pose · whole body · plain background · no panels or props

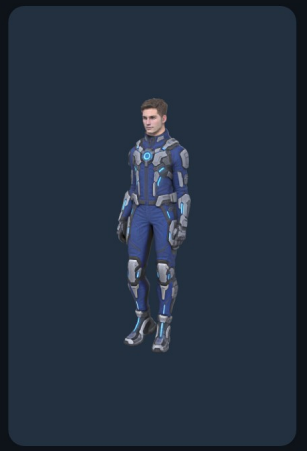
AI rewrites it for Tripo; I approve

4 · Clean image



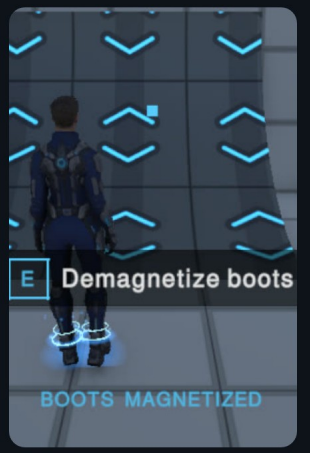
Front, then back from it

5 · Tripo 3D



Model, rig, animations

6 · In Unity



AI fits and wires it

The second pass is the trick: image-to-3D wants one clean view, not a pretty concept sheet.

EXAMPLE 1 · THE PLAYER

A CONCEPT SHEET BECOMES A RIGGED HERO

FIRST TRY · MY PROMPT

“Athletic man in blue jumpsuit with sci-fi exoskeleton, glowing blue circle on chest and back plate. Front and back views. Sci-fi cyberpunk boots with flat gray panels.”

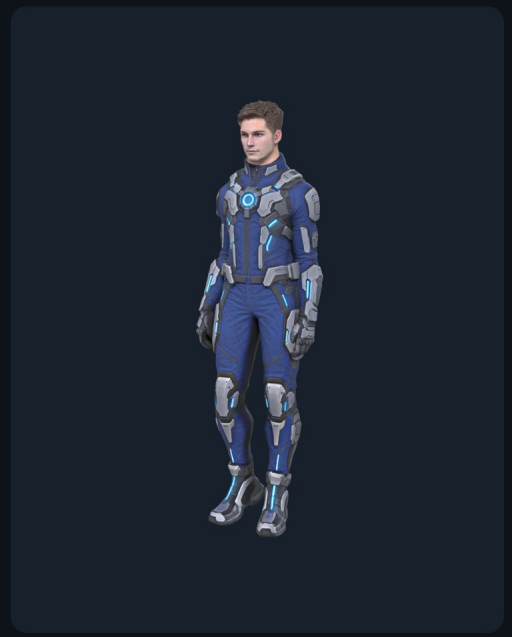


REFINED FOR TRIPO

Front view only, back view made from it · A-pose, arms 45° out · plain grey background, no shadows · slim silhouette, nothing loose



TRIPO MODEL



Humanoid rig; Idle, Walk, Run, Jump, plus four text-to-animation clips

EXAMPLE 2 · THE TUBE ELEVATOR

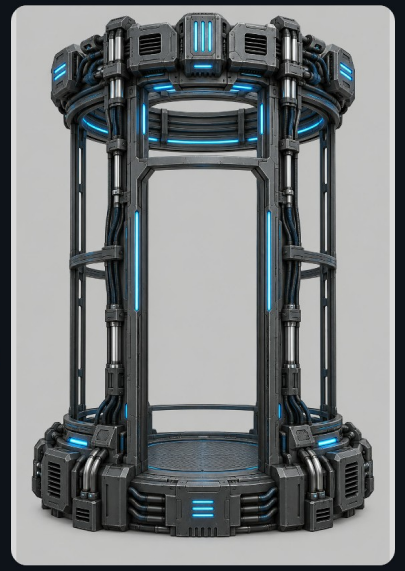
FOUR IMAGES, STEERED BY EYE



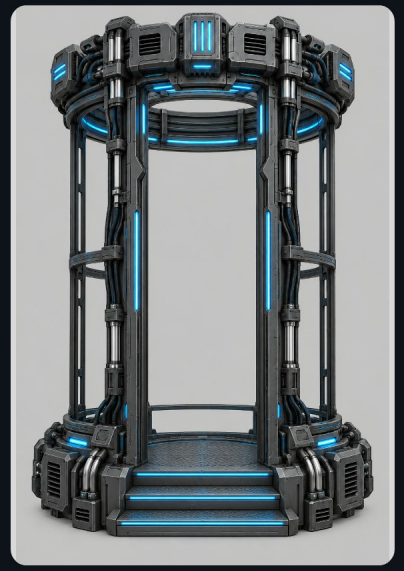
1 · My prompt. Glass tube, machinery at base and top.



2 · Tripo-ready. One view, open frame. “Too plain.”



3 · Heavier. Collar and crown, pipes, protrusions.



4 · Steps added. Then a matching back view.



5 · Tripo model. Its empty floor ring holds our moving platform.

EXAMPLES 3 AND 4

PROMPTING FOR HOW IT WILL BE USED

Sentry

Asked for a domed head that reads as a separate part, so it could turn to aim on its own.



Moving platform

I spotted the thrusters underneath would hit the level, so: a flat slab, detail on the edges only.



INTO THE ENGINE

FROM TRIPO EXPORT TO PLAYABLE

Scale and axes

Exports arrive about 1 m tall and Z-up. The sentry first came in lying on its back.

Split to animate

40 auto-segmented parts grouped into a body and a head that aims.

Clips by playing

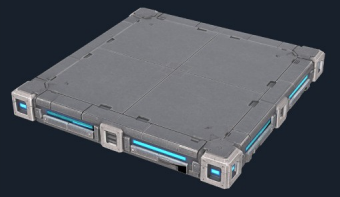
Three text-to-animation walks tried in game; the third, at 3 m/s, stayed.

Skins on boxes

Models stretch over simple collider boxes, so the physics never changed.








SOUND

GENERATED IN ROUNDS, CHOSEN BY EAR

1. The agent writes prompts and calls the ElevenLabs API.

2. Options go on a listening page, matched in loudness.

3. I pick, then we hear it in the game.

4. My playtest notes set the next round.

3 rounds · 31 generations · 13 sounds in the game · 723 credits

CHOSEN FOOTSTEPS, OPTION D OF 4

“...Tactical boots with hard composite soles on a smooth spaceship floor panel, crisp heel and toe”

CHOSEN SENTRY LOCK-ON, OPTION A OF 4

“Menacing robot sentry waking and locking on: a deep low mechanical growl rising slightly, ending in a heavy low targeting thrum”

FOUND BY PLAYTESTING

WHAT THE TESTS COULDN'T SEE



Before: the start shaft flickered on the ride up



After: walls inset 5 mm from the floor's cut edges

Jitter while riding platforms

The player was drawn smoothly between physics steps; platforms weren't. Now both are, and the gap holds at 2.0 cm.

Flickering corridor floors, twice

Floors ran into the doorways. Fixed, then reintroduced by the script that built the new rooms.

FOUND BY PLAYTESTING

WHAT FELT WRONG, AND WHY

What I noticed	The cause	The fix
Sentries beaten by jumping or speed	Reacted only once you were in its plane	Charges early; fires the instant you enter
A descending platform pushed me through the floor	Push-out took the shortest way: down	Step the player aside instead
Footsteps and boot hum barely audible	Two clips silent; hum all below 60 Hz	Recut; new prompts asking for midrange
Footsteps slightly out of time	Fired 0.05–0.1 s before the heel landed	Trigger when the foot stops moving
Never saw the first “magnetize” hint	Hint only looked for curves	Show it on the first magnetic floor
Magnetized walk “too slow to be fun”	1.5 m/s with a heavy walk clip	New Tripo clip at 3 m/s

THE WORKFLOW

HUMAN FIRST, AI IN THE MIDDLE, HUMAN LAST

Human first

The concept and its rules: curves only, drops as puzzles

Every asset idea and first prompt

Which rooms, which hazards, how hard

The models themselves, in Tripo

AI in the middle

Code, tests and levels, in the live Editor

Prompts rewritten for Tripo and ElevenLabs

Diagnosis by measuring, not guessing

Scripted runs proving every room solvable

Human last

Playtest every change

Choose between options: clips, sounds, images

Say what feels wrong

Approve anything risky, then commit

The AI did most of the typing. The decisions about what the game is, and whether it’s good, stayed with me.



Play: aembear.itch.io/mission-apex-strike

Code and prompt log: github.com/williamedwards/mission-apex-strike

THANK YOU